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Topic 13.1: Trapezoidal Rule (Examples)

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Example 1.

Use the trapezoidal rule to approximate the integral of $f(x) = x^3$ on the interval $[1, 2]$.

$$\frac{1}{2}(f(1) + f(2))(2 - 1) = 4.5$$

The actual value of the integral is 3.75 .

Example 2.

Use the trapezoidal rule to approximate the integral of $f(x) = e^{-0.1x}$ on the interval $[2, 5]$.

$$\frac{1}{2}(f(2) + f(5))(5 - 2) = 2.137892120$$

The actual value of the integral is 2.122000934 .

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